

Yatharth Singhal

Curriculum Vitae

Department of Radiation Oncology
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🌐 My Webpage

Research Interests

My research advances **Human-Computer Interaction (HCI)** through multimodal haptic systems that integrate thermal, tactile, and visual feedback for immersive **extended reality (XR)** experiences. I investigate human **perception** and **cognition** to design intuitive interfaces, with particular focus on thermal haptics, perceptual illusions, and adaptive feedback systems. My interdisciplinary approach combines computer science, psychology, and design to create next-generation interactive technologies. My long-term vision centers on leveraging **artificial intelligence (AI)** and **human perception** to advance **XR applications** in clinical, educational, and therapeutic contexts, developing more adaptive, natural, and engaging digital interactions.

Keywords: *Human-Computer Interaction, Haptic Technologies, Extended Reality, Multimodal Feedback, Artificial Intelligence, Human Perception, Medical Applications*

Education

- 2021 – 2025 **Doctor of Philosophy in Computer Science**, *The University of Texas at Dallas*, Texas, USA.
Dissertation: “*Perceptual modulation in XR: Designing thermal and wet feedback through illusions.*”
Advisor: Dr. Jin Ryong Kim
GPA: 3.81/4.0
- 2019 – 2021 **Master of Science in Computer Science**, *The University of Texas at Dallas*, Texas, USA.
GPA: 3.83/4.0
- 2014 – 2018 **Bachelor of Technology in Computer Science**, *Jaypee Institute of Information Technology*, Noida, India.
GPA: 3.75/4.0

Professional Experience

Academic Positions

- Sep 2025 – Present **Instructor, Department of Radiation Oncology**, *UT Southwestern Medical Center*, Dallas, TX.
- Develop and maintain clinical software tools that support radiation oncologists, physicists, and clinical staff in their daily workflows.
 - Improve and update existing applications by implementing new features, fixing bugs, and ensuring reliable performance within the clinical environment.
- Sep 2020 – Aug 2025 **Research Assistant**, *University of Texas at Dallas*, Richardson, TX.
- Developed multimodal haptic interfaces integrating thermal and tactile feedback, employing perceptual illusions (thermal referral, thermal masking, wetness illusion) to advance realism and immersion in VR.
 - Designed and executed controlled user studies evaluating perception, usability, and interaction quality in virtual environments with multimodal feedback.
 - Proposed and implemented optimization frameworks for thermal–tactile interaction, enhancing performance, efficiency, and user experience in haptic systems.

Industry Experience

- Jun 2018 – Jun 2019 **Systems Engineer**, *Infosys*, Bhubaneswar, India.
- Earned accelerated promotion by completing advanced technical training with top-tier performance.
 - Designed and deployed analytics dashboards for healthcare and banking clients, supporting large-scale data-driven decision-making.

Awards and Honors

- 2025 **Best Paper Honorable Mention Award** at ACM UIST 2025 for *HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements* (Top 5%)
- 2025 **Best Demo Award** at IEEE WHC 2025 for *Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction*
- 2024 **Outstanding Student Research Award**, The University of Texas at Dallas
- 2024 **Best Demonstration Award** at IEEE ISMAR 2024 for *Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction*
- 2024 **Best Demo Honorable Mention Award** at ACM UIST 2024 for *Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction*
- 2024 **Best Demo Honorable Mention Award** at IEEE Haptic Symposium 2024 for *Demonstrating Upper Body Thermal Referral and Tactile Masking for Localized Feedback*
- 2023 **Best Student Demonstration Award** at SIGGRAPH Asia 2023 for *FIRE: Mid-Air Thermo-Tactile Display*

Publications

* indicates co-first authorship. Citation count: 230+ (Google Scholar)

Journal Articles

- J2 Haokun Wang, **Yatharth Singhal**, and Jin Ryong Kim, *Let It Snow: Designing Snowfall Experience in VR*, Proceedings of ACM Interactive Mobile Wearable Ubiquitous Technology (IMWUT) Vol. 8, Iss. 2, Article 54, June 2024 [**Impact Factor: 4.5**].
- J1 Hyunki Son*, **Yatharth Singhal***, Haokun Wang, and Jin Ryong Kim, *Upper Body Thermal Referral and Tactile Masking for Localized Feedback*, IEEE Transactions on Visualization and Computer Graphics (TVCG) [**Impact Factor: 6.5**] **Best Paper Nomination**.

Peer-Reviewed Conference Papers

- C10 **Yatharth Singhal, Abbas Khawaja, and Jin Ryong Kim**, *Moisture Transfer: A Perceptual Wetness Illusion Through Thermal and Wet Integration*, In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2026) [**Acceptance Rate: 25.3%**].
- C9 Ayush Bhardwaj, Ashish Pratap, Abbas Khawaja, **Yatharth Singhal**, Hyunjae Gil, and Jin Ryong Kim, *Understanding Latency Sensitivity in Thermal and Tactile Feedback for Multimodal Haptics in VR*, In Proceedings of the 31st ACM Symposium on Virtual Reality Software and Technology (VRST 2025) [**Acceptance Rate: 27.0%**].
- C8 **Yatharth Singhal**, Daniel Honrales, and Jin Ryong Kim, *HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements*, In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST 2025) [**Acceptance Rate: 22.2%**] **Best Paper Honorable Mention Award**.
- C7 **Yatharth Singhal**, Daniel Honrales, Haokun Wang and Jin Ryong Kim, *Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction*, In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST 2024) [**Acceptance Rate: 24.0%**].
- C6 Haokun Wang, **Yatharth Singhal**, Hyunjae Gil, and Jin Ryong Kim, *Fiery Hands: Designing Thermal Glove through Thermal and Tactile Integration for Virtual Object Manipulation*, In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST 2024) [**Acceptance Rate: 24.0%**].
- C5 **Yatharth Singhal**, Daniel Honrales, Hsin-Ni Ho, and Jin Ryong Kim, *Wetness Illusion in Mid-Air*, IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024) [**Acceptance Rate: 30.4%**].

- C4 Haokun Wang, **Yatharth Singhal**, Hyunjae Gil, and Jin Ryong Kim, *Thermal Masking: When the Illusion Takes Over the Real*, In Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI 2024**) [**Acceptance Rate: 26.3%**].
- C3 Haokun Wang, **Yatharth Singhal**, and Jin Ryong Kim, *Fabric Thermal Display using Ultrasonic Waves*, IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2023**) [**Acceptance Rate: 27.5%**].
- C2 **Yatharth Singhal**, Richard Noeske, Ayush Bhardwaj, and Jin Ryong Kim, *Improving Finger Stroke Recognition Rate for Eyes-Free Mid-Air Typing in VR*, In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (**CHI 2022**) [**Acceptance Rate: 24.7%**].
- C1 **Yatharth Singhal**, Haokun Wang, Hyunjae Gil, and Jin Ryong Kim, *Mid-air thermo-tactile feedback using ultrasound haptic display*, In Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology (**VRST 2021**) [**Acceptance Rate: 25.7%**].

Conference Demonstrations and Posters

- D6 **Yatharth Singhal**, Daniel Honrales, Haokun Wang, and Jin Ryong Kim, *Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction*, IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**) **Best Demo Award**.
- D5 **Yatharth Singhal**, Haokun Wang, and Jin Ryong Kim, *Demonstration of FIRE: Mid-Air Thermo-Tactile Display*, IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**).
- D4 Haokun Wang, **Yatharth Singhal**, Hyunjae Gil, and Jin Ryong Kim, *Demonstration of Fiery Hands: Thermal Gloves through Thermal and Tactile Integration*, IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**).
- D3 **Yatharth Singhal**, Haokun Wang, and Jin Ryong Kim, *FIRE: Mid-Air Thermo-Tactile Display*, In ACM SIGGRAPH 2024 Emerging Technologies (**SIGGRAPH 2024**).
- D2 **Yatharth Singhal**, Haokun Wang, and Jin Ryong Kim, *Demonstrating Mid-Air Ultrasound Haptics with Thermal Display*, IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (**VRW 2024**).
- D1 **Yatharth Singhal**, Haokun Wang, and Jin Ryong Kim, *FIRE: Mid-Air Thermo-Tactile Display*, In SIGGRAPH Asia 2023 Emerging Technologies (**SIGGRAPH ASIA 2023**) **Best Demo Award**.

Professional Service and Reviewing

Conference Organization

- 2026 **Program Committee Member**, ACM TEI 2026, Chicago, IL.
Program committee member for the Work In Progress track

Editorial and Review Activities

- 2025-Present **Reviewer:** ACM UIST, ACM IMWUT
- 2024-Present **Reviewer:** ACM CHI, IEEE TOH, IEEE TVCG, IEEE VR, IEEE Haptic Symposium, AsiaHaptics, Eurohaptics
- 2023-Present **Reviewer:** IEEE ISMAR, ACM ICMI, IEEE WHC, ACM SIGGRAPH

Teaching Experience

Teaching Assistant (UT Dallas)

Graduate Courses: CS6334 Virtual Reality (4 semesters), CS6326 Human-Computer Interactions (1 semester)

Undergraduate Courses: CS4352 Human-Computer Interactions (2 semesters), CS4361 Computer Graphics (1 semester), CS4392 Computer Animation (1 semester), CS4384 Automata Theory (2 semesters), CS4347 Database Management (2 semesters)

Student Evaluation Scores: Average 4.6/5.0 across all courses taught

Guest Lectures

- 2024 **"Haptic Technologies in Virtual Reality"**, *CS6334: Virtual Reality*, UT Dallas.
Delivered a comprehensive lecture on haptic feedback systems and their applications in VR
- 2023 **"Thermal Interfaces and Human Perception"**, *CS6326: Human-Computer Interactions*, UT Dallas.
Introduced students to multimodal interface design and perceptual psychology principles

Research and Technical Skills

Research Methodologies

Experimental Design, Statistical Hypothesis Testing, Psychophysical Methods, User-Centered Design, Qualitative and Quantitative Analysis, Usability Testing, Heuristic Evaluation, Interview and Survey Design

Programming and Development

Python, C/C++, C#, Java, JavaScript, Unity3D, Unreal Engine, Arduino/Raspberry Pi, Machine Learning (PyTorch, TensorFlow, scikit-learn), Computer Vision (OpenCV), Natural Language Processing, Web Development (React, Node.js, Angular)

Statistical Analysis and Data Science

R, MATLAB, SPSS, JASP, Advanced Statistical Methods (ANOVA, GLMMs, Bayesian Analysis), Data Visualization, Signal Processing

Hardware Prototyping and Design

3D Printing and Modeling (Fusion 360, Blender), PCB Design (KiCad), Sensor Integration, Thermal Control Systems, Ultrasonic Haptic Devices, UI/UX Design (Figma, Adobe Creative Suite)

Mentoring and Outreach

Student Mentoring

- 2021 - 2025 **Lab Training Program Coordinator**, *Multimedia Interaction Lab, UT Dallas*.
Designed and conducted comprehensive training programs for 30+ graduate (MS/PhD) and undergraduate students in VR/AR development, haptic technologies, experimental design, and research methodologies
- 2022 - 2024 **Lead Mentor, UTD CS Summer Research Program**, *UT Dallas*.
Designed and led hands-on VR and haptics research experiences for 25+ high school students over three summers. Developed curriculum integrating computer science theory with practical applications.
- 2022 **Lead Mentor, UTD STEM Bridge Summer Camp**, *UT Dallas*.
Mentored high school students in Unity 3D development and VR applications. Team achieved 3rd place in the competitive STEM Bridge Competition.

Direct Student Mentees

- 2023-2025 **Daniel Honrales** (Undergraduate Student, UT Dallas CS) | *Now: Ph.D. at UT Dallas*
- 2023-2025 **Shreya Krishnan** (Undergraduate Student, UT Dallas CS) | *Now: UC Berkley*
- 2022 - 2025 **Varun Sridhar** (High School Student, Plano East) | *Now: University of Pennsylvania*
- 2023 **Kunal Tiwari** (High School Student, Independence) | *Now: UT Austin*

Professional Memberships

- 2021-Present **Association for Computing Machinery (ACM)** - Student Member
- 2021-Present **IEEE Computer Society** - Student Member
- 2022-Present **ACM Special Interest Group on Computer-Human Interaction (SIGCHI)** - Student Member